SIMATS ENGINEERING
CO5-ASSESSMENT TOOL3-Reverse Engineering

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1709**

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COURSE OUTCOME COVERED

CO5: *Design intelligent agents and real-time AI-based systems (chatbots, expert systems, emotion detection, edge AI, autonomous drones, and related applications) by selecting appropriate agent architectures, knowledge representation, and reasoning mechanisms to solve real-world problems. (BL6)*

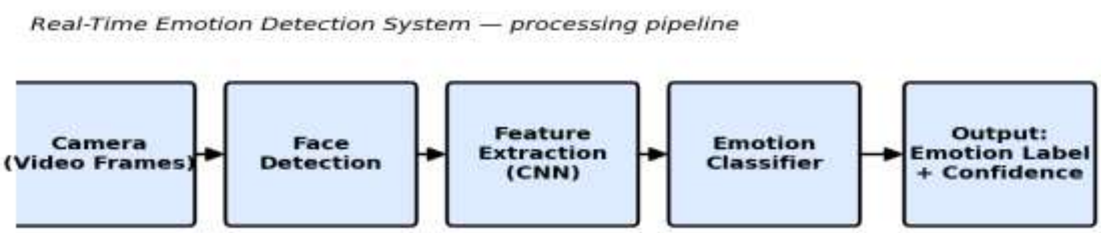
Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks:50**

Q1. The following is a conversation between a user and an AI-based flight-booking chatbot. Study the transcript given below and reverse-engineer the chatbot's design by: (a) writing its PEAS description, (b) classifying the type of agent it represents with justification, (c) identifying the likely component (NLU, Dialogue Manager, Knowledge Base, or NLG) responsible for each chatbot response, and (d) suggesting one enhancement using a learning-agent architecture.

Speaker	**Message**
User	I want to book a flight to Chennai next Friday.
Chatbot	Sure! What time would you like to depart, and do you have a preferred airline?
User	Morning flight, no preference.
Chatbot	I found 3 morning flights to Chennai on Friday. Shall I show you the cheapest option first?

Q2. The block diagram below shows the processing pipeline of a Real-Time Emotion Detection System. Reverse-engineer the system by: (a) explaining the function performed at each stage of the pipeline, (b) identifying the likely AI/ML technique used at each stage, (c) writing the PEAS description of the system, and (d) classifying the agent type (e.g., simple reflex, model-



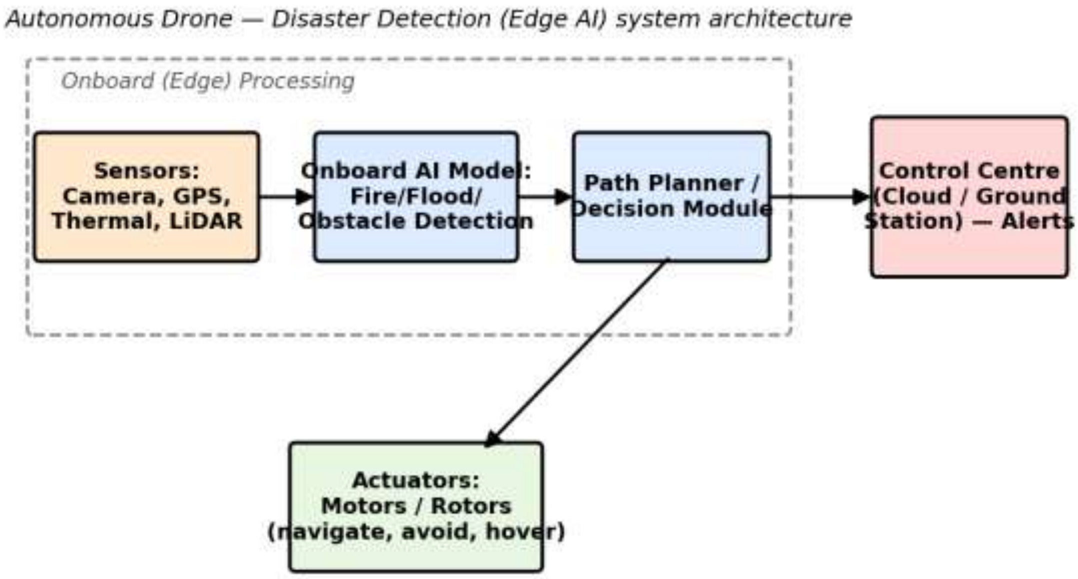
based) with justification.

Q3. An Expert System Shell for medical diagnosis contains the rule base, input facts, and reasoning trace given below. Study it and reverse-engineer the shell by: (a) identifying the type of chaining (forward/backward) used to reach the diagnosis, with justification, (b) identifying the expert-system-shell components illustrated (knowledge base, inference engine, explanation facility), (c) explaining how the shell would perform knowledge acquisition to add a new rule for dengue, and (d) explaining, with reasoning, whether backward chaining could reach the same diagnosis.

Rule	IF (condition)	THEN (conclusion)
R1	fever = yes AND cough = yes	suspect = flu
R2	suspect = flu AND body_ache = yes	diagnosis = Influenza
R3	fever = yes AND rash = yes	suspect = measles

Given facts: fever = yes, cough = yes, body_ache = yes
Reasoning trace: R1 fires first (fever & cough matched) → suspect = flu. R2 then fires (suspect = flu & body_ache matched) → diagnosis = Influenza.
Explanation generated by the shell: *"Diagnosis = Influenza because fever and cough indicated flu, and flu together with body ache confirmed Influenza."*

Q4. The diagram below shows the system architecture of an Autonomous Drone used for Disaster Detection with Edge AI deployment. Reverse-engineer the design by: (a) writing the PEAS description of the drone agent, (b) classifying the agent type (goal-based/utility-based) with justification, (c) identifying which components run on the edge versus the cloud and justifying why on-edge processing was chosen for detection and obstacle avoidance, and (d) identifying whether the agent's control architecture is reactive, deliberative, or hybrid, with justification.



Q5. The table below shows sample output logs from an AI-based Deepfake & Face Authenticity Verification system across five video frames. Study the output and reverse-engineer the system by: (a) identifying the likely input features used by the model (e.g., blink rate, texture/artifact score), (b) identifying the type of model/technique likely used to produce these outputs, (c) explaining how the final Real/Fake decision may have been derived from the frame-level confidence scores, and (d) suggesting one way a swarm/ensemble-based approach could improve the system's reliability.

Frame No.	Blink Rate (per 10s)	Texture/Artifact Score	Fake Confidence (%)	Prediction
1	4	0.12	8	Real
2	1	0.61	74	Fake
3	0	0.83	91	Fake
4	5	0.09	5	Real
5	2	0.55	68	Fake

Overall system verdict: Fake (3 of 5 frames flagged — majority vote)
Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Max Marks
Identification of Agent Type & PEAS (correctly inferred from the given artifact)	Correctly identifies the agent type and gives a complete, accurate PEAS description, fully consistent with the given artifact.	Identifies the agent type and PEAS description correctly with only minor omissions.	Partially correct identification of agent type/PEAS; some elements are missing or inconsistent with the artifact.	Agent type/PEAS identification is largely incorrect or not attempted.	**10**
Identification of Architecture & Components (mapping artifact evidence to system components)	Accurately identifies all relevant architectural components and correctly maps each to specific evidence in the artifact.	Identifies most components correctly with reasonable mapping to the artifact; minor gaps present.	Identifies some components; mapping to the artifact is weak or partially incorrect.	Little or no correct identification of architecture/components.	**10**
Identification of Underlying Technique/Algorithm/Reasoning Mechanism	Correctly reverse-engineers the technique, algorithm, or reasoning mechanism (e.g., chaining type, ML	Correctly identifies the technique/mechanism with adequate, mostly	Identification is only partially correct or lacks sufficient supporting evidence.	Technique/mechanism is incorrectly identified or not addressed.	**10**

Criteria	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Max Marks
	technique) with clear supporting evidence.	accurate justification.			
Quality of Justification & Use of Evidence (linking conclusions back to the given artifact)	Every conclusion is clearly justified with specific, accurate references to the given artifact.	Most conclusions are justified with reasonable references to the artifact.	Justification is vague or only loosely connected to the artifact.	Conclusions are stated with little or no justification or evidence.	**10**
Clarity & Completeness of Presentation (addresses all sub-parts of the question in an organised manner)	All sub-parts are addressed completely, in a clear, well-organised, and logically structured manner.	Most sub-parts are addressed clearly with minor gaps in organisation or completeness.	Some sub-parts are missing or the presentation is unclear/poorly organised.	Response is incomplete, disorganised, or largely missing.	**10**
				Total	50

Q1: Flight-Booking Chatbot Analysis

(a) PEAS Description

- **Performance Measure:** Successful flight booking, user satisfaction, finding the cheapest/best flight, and conversational efficiency.
- **Environment:** The chat interface, user text inputs, and the external airline flight database.
- **Actuators:** The screen display outputting text to the user, and the backend booking system API.
- **Sensors:** The text input box receiving the user's message and the database query response system.

(b) Agent Type Classification

- **Classification:** Utility-based Agent.
- **Justification:** While a simple goal-based agent would just book *any* available flight to Chennai to achieve the core objective, this chatbot explicitly attempts to maximize user satisfaction. By offering to "show you the cheapest option first," it proves it is evaluating multiple paths (the 3 morning flights) and ranking them based on a utility function (price) to provide the optimal choice.

(c) Component Identification

- **Response 1 ("What time... preferred airline?"): Dialogue Manager (DM)** and **Natural Language Generation (NLG)**. After the Natural Language Understanding

(NLU) component processed the initial request, the DM recognized that mandatory information slots (time, airline) were still empty. It then instructed the NLG to generate the follow-up question.
- **Response 2 ("I found 3 morning flights..."): Knowledge Base (KB)**. The system used the newly extracted parameters (morning, no preference) to query the KB. The KB successfully retrieved the three matching flight records, which were then passed back to the DM and NLG to formulate the final response.

(d) Learning-Agent Enhancement

- **Enhancement:** Introduce a user-profile learning module. The *Critic* component could monitor the user's selections over multiple bookings. If the user consistently picks the cheapest morning flights, the *Learning Element* updates the system so the *Performance Element* can proactively suggest budget morning flights in future sessions, skipping the redundant slot-filling questions entirely.

Q2: Real-Time Emotion Detection System Analysis

(a) Function of Each Pipeline Stage

- **Camera (Video Frames):** Ingests continuous live video and standardizes it into a sequence of individual image frames for processing.
- **Face Detection:** Scans the raw video frame to locate human faces, drawing a bounding box around the face and cropping out irrelevant background data.
- **Feature Extraction (CNN):** Analyzes the isolated face image to detect complex geometric patterns, such as the curve of the mouth, the wrinkling of the nose, or the distance between the eyebrows.
- **Emotion Classifier:** Maps the extracted complex features against learned patterns to determine which emotional category they best represent.
- **Output (Label + Confidence):** Generates the final decision (e.g., "Happy", "Angry") along with a statistical probability (e.g., 95%) of the prediction's accuracy.

(b) Likely AI/ML Techniques Used

- **Face Detection:** Often utilizes algorithms like Haar Cascades, HOG (Histogram of Oriented Gradients), or MTCNN (Multi-task Cascaded Convolutional Networks).
- **Feature Extraction:** As specified in the diagram, this relies on a **Convolutional Neural Network (CNN)**, which uses mathematical convolution layers and pooling to automatically learn spatial hierarchies of features.
- **Emotion Classifier:** Typically uses a Fully Connected (Dense) Neural Network layer ending with a Softmax activation function to output a probability distribution across various emotion classes.

(c) PEAS Description

- **Performance Measure:** High accuracy in emotion classification, minimal processing latency (real-time speed), and robustness to varying lighting conditions or facial angles.
- **Environment:** The physical space containing the human subject (e.g., a user sitting in front of a webcam, a driver in a vehicle cabin, or a retail store aisle).
- **Actuators:** The display interface or API endpoint that outputs the final emotion label and confidence score.
- **Sensors:** The optical camera lens capturing the video frames.

(d) Agent Type Classification

- **Classification:** Simple Reflex Agent.
- **Justification:** The pipeline described processes each video frame individually to generate an immediate output. It operates entirely on the *current percept* (the current video frame) and applies a pre-learned mapping (the trained weights of the CNN and Classifier) to trigger an immediate action (outputting the label). It does not require memory of past frames or evaluate future states to make its decision, which perfectly aligns with a simple condition-action rule architecture.

Q3: Medical Diagnosis Expert System Analysis

(a) Type of Chaining Used

- **Classification:** Forward Chaining.
- **Justification:** The provided reasoning trace explicitly begins with a set of known input facts (fever = yes, cough = yes, body_ache = yes). The system then looks for rules whose conditions match these facts, firing R1 first to infer a new fact (suspect = flu), which then allows R2 to fire and reach the final conclusion (diagnosis = Influenza). Moving systematically from initial facts to a final conclusion is the exact definition of data-driven forward chaining.

(b) Expert-System-Shell Components Illustrated

- **Knowledge Base:** Represented by the explicit list of conditional rules (R1, R2, and R3) containing the medical logic.
- **Inference Engine:** Demonstrated by the "Reasoning trace," which is the active processing component that successfully matched the input facts against the conditions of R1 and R2 to deduce the result.
- **Explanation Facility:** Shown by the human-readable output text ("Diagnosis = Influenza because...") that translates the execution trace into an understandable justification for the user.

(c) Knowledge Acquisition for a New Rule

- To add a rule for dengue, a domain expert or knowledge engineer would use the shell's **Knowledge Acquisition Facility** (often a graphical interface or specialized rule-entry editor). The expert would define the specific IF-THEN logic—such as IF fever = yes AND joint_pain = yes THEN suspect = dengue—and the facility would compile this logic, check for conflicts with existing rules, and seamlessly integrate it into the active Knowledge Base.

(d) Backward Chaining Applicability

- **Explanation:** Yes, backward chaining could successfully reach the exact same diagnosis.
- **Reasoning:** Backward chaining is goal-driven. The system would start with the hypothesis diagnosis = Influenza. To prove this, it would check R2 and realize it needs to verify suspect = flu and body_ache = yes. Since body_ache is already a given fact, it then sets a new sub-goal to prove suspect = flu. It looks at R1 and asks if fever = yes and cough = yes. Because both of these are provided as given facts, the sub-goal is met, proving the main hypothesis true.

Q4: Autonomous Drone Disaster Detection System Analysis

(a) PEAS Description

- **Performance Measure:** Disaster detection accuracy (fire/flood), low latency in obstacle avoidance, flight stability, battery efficiency, and reliable alert delivery.
- **Environment:** Dynamic disaster zones with poor connectivity, physical obstacles, smoke, fire, and changing weather conditions.
- **Actuators:** Motors and rotors (used to navigate, avoid obstacles, and hover).
- **Sensors:** Camera, GPS, Thermal imaging sensor, and LiDAR.

(b) Agent Type Classification

- **Classification:** Goal-based Agent (or Utility-based Agent if path optimization is emphasized).
- **Justification:** The system continuously evaluates sensory inputs to fulfill specific mission goals—detecting disaster hazards (fire/flood) and navigating safe flight paths toward waypoints without collisions. It selects actions based on desirable future outcomes rather than merely executing direct stimulus-response mappings.

(c) Edge vs. Cloud Distribution & On-Edge Justification

- **Onboard (Edge) Components:** Sensors (Camera, GPS, Thermal, LiDAR), Onboard AI Model (Fire/Flood/Obstacle Detection), and the Path Planner / Decision Module.
- **Cloud / Ground Station Components:** Control Centre for receiving high-level alerts and global mission coordination.
- **Justification for On-Edge Processing:**
    - **Ultra-Low Latency:** Obstacle avoidance and flight stabilization require sub-millisecond reactions to prevent crashes.
    - **Network Independence:** Disaster areas frequently suffer from damaged or non-existent cellular/Wi-Fi infrastructure, necessitating autonomous local decision-making.

(d) Control Architecture Classification

- **Classification:** Hybrid Control Architecture.
- **Justification:** The drone incorporates both **reactive** behaviors (rapid, low-level reflexes for obstacle avoidance and rotor adjustments to keep the drone airborne) and **deliberative** behaviors (high-level path planning, navigation toward target coordinates, and structured disaster assessment).

Q5: Deepfake & Face Authenticity Verification System Analysis

(a) Likely Input Features

- Based on the provided output logs, the system explicitly extracts and relies on temporal and spatial features, specifically the **Blink Rate (per 10s)** and the **Texture/Artifact Score**.
- *Justification:* Deepfakes often struggle with physiological micro-expressions (like natural blinking) and introduce spatial inconsistencies (blurring or unnatural pixel blending), which these specific features are designed to capture.

(b) Type of Model/Technique Used

- **Classification:** A hybrid spatial-temporal Deep Learning approach, likely a Convolutional Neural Network (CNN) combined with a Recurrent Neural Network (RNN) or Long Short-Term Memory (LSTM) network.
- *Justification:* The "Texture/Artifact Score" suggests a CNN is being used to evaluate the spatial, frame-by-frame image quality. The "Blink Rate" requires analyzing movement across multiple frames over time, which points toward an RNN/LSTM or a temporal heuristic tracker evaluating a sequence of frames.

(c) Derivation of Final Real/Fake Decision

- **Frame-Level Decision:** The system likely uses a confidence threshold (e.g., 50%) applied to the "Fake Confidence (%)". For instance, Frames 2, 3, and 5 have scores well above 50% (74%, 91%, 68%) and are labeled "Fake". Frames 1 and 4 have low scores (8%, 5%) and are labeled "Real".
- **Overall System Verdict:** To prevent a single glitchy frame from ruining the result, the system aggregates the individual frame predictions over a temporal window (5 frames) and applies a **majority vote** logic. Since 3 out of the 5 frames were flagged as fake, the final output is Fake.

(d) Swarm/Ensemble-Based Improvement

- **Enhancement:** The system's reliability could be significantly improved by deploying a heterogeneous ensemble architecture. Instead of relying on a single monolithic model to score the frame, the system could run multiple specialised, independent agents concurrently—for example, one agent exclusively analyzing lip-syncing anomalies, a second agent checking frequency-domain artifacts, and a third tracking eye-blinking. A final "meta-classifier" or swarm-voting mechanism would then aggregate these diverse agent predictions to reach a highly robust final verdict, drastically reducing false positives against sophisticated deepfakes.